Intel RealSense Depth Camera D455: Twice the Range, Better Performance

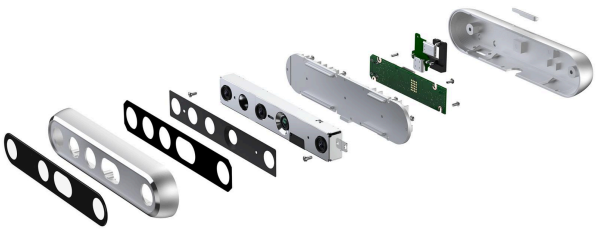
Intel introduces the Intel RealSense Depth Camera D455, a new stereo-based depth camera equipped with a longer range and increased precision.

News

- June 16, 2020
- Contact Intel PR

More New Technologies News

The Intel RealSense Depth Camera D455 increases the optimal range to 6 meters without sacrificing on field of view. The D455 includes global shutters for the depth and RGB sensors to improve correspondence between the two different data streams and to match the field of view between the depth sensors and the RGB sensor. Intel Corporation introduced the camera on June 16, 2020. (Credit: Intel Corporation)



The Intel RealSense Depth Camera D455 increases the optimal range to 6 meters without sacrificing on field of view. The D455 includes global shutters for the depth and RGB sensors to improve correspondence between the two

different data streams and to match the field of view between the depth sensors and the RGB sensor. Intel Corporation introduced the camera on June 16, 2020. (Credit: Intel Corporation)

The Intel RealSense Depth Camera D455 is a new stereo-based depth device that achieves twice the performance of previous generation models. The D455 provides developers with longer range and accuracy to design across a wide variety of indoor and outdoor applications. Intel Corporation introduced the camera on June 16, 2020. (Credit: Intel Corporation)

» Download all images (ZIP, 7 MB)

What's New: Today, Intel introduced the Intel® RealSense™ Depth Camera D455, a new stereo-based depth camera equipped with a longer range and increased precision that achieves twice the range of previous generation models. The D455 grants developers the freedom to design vision-based solutions that give devices the ability to see, understand, interact with and learn from their environments.

> *"The Intel RealSense family of products now offers developers more choices. The D455 expands the capabilities of our stereo camera line allowing us to address a wider range of applications with longer range use cases and allowing users to design solutions that best meet their needs."*
>
> –Sagi BenMoshe, Intel corporate vice president and general manager of the Intel RealSense Group

How It's Different: Stereo depth works in a wide variety of lighting conditions. With its long range, enhanced accuracy and easy deployment, the

D455 camera is a powerful tool that can be used for both indoor and outdoor applications.

For example, a key requirement for autonomous movement of a robot is the ability to “see” its environment. The longer range and improved coverage provided by the D455 allows a robot to scan its environment and make critical choices, like obstacle avoidance.

In healthcare applications, by mounting the D455 in the corner of a hospital room, staff can closely monitor patients through alerts based on movement via third-party software. This enhances patient care substantially as it frees up critical time for healthcare workers to tend to other priorities.

Why It’s Important: The D455 camera increases the optimal range to 6 meters, making it twice as accurate as the current D400 cameras without sacrificing field of view. The D455 also includes global shutters for the depth and RGB sensors to improve correspondence between the two different data streams and to match the field of view between the depth sensors and the RGB sensor. In addition, this camera also integrates an IMU to allow for refinement of its depth awareness in any situation where the camera moves.

How It Works: The D455 achieves less than 2% Z-error at 4 meters with several improvements. First, the depth sensors are located 95 millimeters apart, providing greater depth accuracy at a longer range. Second, the depth and RGB sensors are placed on the same stiffener, resulting in an improved alignment of color and depth. Lastly, the RGB sensor has the same field of view as the depth sensors, further improving correlation of depth and color points.

The D455 utilizes the same SDK 2.0 as the entire portfolio of Intel RealSense devices to leverage existing code base, allowing for easy upgrades from any of Intel’s depth cameras with virtually no development time.

When You Can Get It: The Intel RealSense Depth Camera D455 is priced at $239 and available for pre-order today. For more details and to order your camera, visit intelrealsense.com

.

More Context: Intel RealSense Technology

(Press Kit)

The Small Print:

[1]disclaimers

Tags

New Technologies

About Intel

Intel (Nasdaq: INTC) is an industry leader, creating world-changing technology that enables global progress and enriches lives. Inspired by Moore's Law, we continuously work to advance the design and manufacturing of semiconductors to help address our customers' greatest challenges. By embedding intelligence in the cloud, network, edge and every kind of computing device, we unleash the potential of data to transform business and society for the better. To learn more about Intel's innovations, go to newsroom.intel.com and intel.com .

© Intel Corporation. Intel, the Intel logo and other Intel marks are trademarks of Intel Corporation or its subsidiaries. Other names and brands may be claimed as the property of others.

Supply Chain Transparency

Site Map

Recycling

Intel technologies may require enabled hardware, software or service activation. // No product or component can be absolutely secure. // Your costs and results may vary. // Performance varies by use, configuration and other factors. // See our complete legal Notices and Disclaimers . // Intel is committed to respecting human rights and avoiding causing or contributing to adverse impacts on human rights. See Intel's Global Human Rights Principles . Intel's products and software are intended only to be used in applications that do not cause or contribute to adverse impacts on human rights.